

MACHINE LEARNING BASICS

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Homework 2

Question 1: Line Search Optimization Problem

- (a) Write the general form of the optimization problem that line-search algorithms solve, and briefly explain the roles of the **search direction** and the **step length**.
- (b) Define a **descent direction** in the context of line search. State the necessary condition for a vector p to be a descent direction at x .
- (c) List at least three common methods to determine the step length α in line-search algorithms. For each method give a short explanation (e.g., exact line search, Armijo/backtracking, Wolfe conditions, fixed step, Barzilai–Borwein).
- (d) Consider the quadratic function

$$f(x_1, x_2) = 2x_1^2 + x_1x_2 + 4x_2^2.$$

At iteration t , the current point and search direction are

$$x^{(t)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad p^{(t)} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

- 1. Show that $p^{(t)}$ is a descent direction at $x^{(t)}$, i.e. show that $\nabla f(x^{(t)})^\top p^{(t)} < 0$.
- 2. Form the one-dimensional function $\phi(\alpha) = f(x^{(t)} + \alpha p^{(t)})$ and simplify it to exhibit its quadratic form.
- 3. Compute $\alpha^* > 0$ that minimizes $\phi(\alpha)$ (exact line search). State whether $\alpha^* > 0$.

Answer:

- (a) **General form and roles.**

The unconstrained problem is

$$\min_{x \in \mathbb{R}^n} f(x).$$

Iteration rule:

$$x^{(t+1)} = x^{(t)} + \alpha^{(t)} p^{(t)}.$$

- (b) $p^{(t)}$ — the *search direction* (where to move).
- (c) $\alpha^{(t)} > 0$ — the *step length* (how far to move).
- (d) If $p^{(t)}$ is a descent direction, small positive $\alpha^{(t)}$ reduces f .

- (e) **Descent direction.**

Criterion: $\nabla f(x)^\top p < 0$.

Consequence: for sufficiently small $\alpha > 0$, $f(x + \alpha p) < f(x)$.

- (f) **Common step-length selection methods.**

- (g) Exact line search $\rightsquigarrow \alpha^* = \arg \min_{\alpha > 0} f(x + \alpha p)$. Optimal along p but potentially expensive.
- (h) Backtracking (Armijo) $\rightsquigarrow$ shrink α (e.g. $\alpha \leftarrow \tau \alpha$) until

$$f(x + \alpha p) \leq f(x) + c\alpha \nabla f(x)^\top p.$$

- (i) Wolfe conditions $\rightsquigarrow$ satisfy Armijo plus curvature:

$$\nabla f(x + \alpha p)^\top p \geq c_2 \nabla f(x)^\top p.$$

- (j) Fixed/scheduled step $\rightsquigarrow$ preset $\alpha^{(t)}$ (constant or decaying).

- (k) Barzilai–Borwein (BB) $\rightsquigarrow$ adaptive scalar from secant ideas, e.g.

$$\alpha_{\text{BB}} = \frac{s^\top s}{s^\top y} \quad \text{or} \quad \frac{s^\top y}{y^\top y},$$

where $s = x^{(t)} - x^{(t-1)}$, $y = \nabla f(x^{(t)}) - \nabla f(x^{(t-1)})$.

- (l) **Quadratic example.**

Given

$$f(x_1, x_2) = 2x_1^2 + x_1x_2 + 4x_2^2, \quad x^{(t)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad p^{(t)} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

1. **Descent check.**

$\nabla f(x) = \begin{pmatrix} 4x_1 + x_2 \\ x_1 + 8x_2 \end{pmatrix}$. At $x^{(t)} = (0, 1)^\top$: $\nabla f(x^{(t)}) = (1, 8)^\top$. Then

$$\nabla f(x^{(t)})^\top p^{(t)} = 1 \cdot 1 + 8 \cdot (-2) = -15 < 0,$$

so $p^{(t)}$ is a descent direction.

2. **One-dimensional function.**

$x^{(t)} + \alpha p^{(t)} = (\alpha, 1 - 2\alpha)^\top$. Hence

$$\begin{aligned} \phi(\alpha) &= f(\alpha, 1 - 2\alpha) = 2\alpha^2 + \alpha(1 - 2\alpha) + 4(1 - 2\alpha)^2 \\ &= 16\alpha^2 - 15\alpha + 4, \end{aligned}$$

a quadratic in α with leading coefficient $16 > 0$.

3. **Exact minimizer.**

For $\phi(\alpha) = A\alpha^2 + B\alpha + C$, $\alpha^* = -B/(2A)$. With $A = 16$, $B = -15$:

$$\alpha^* = \frac{15}{32} = 0.46875 > 0.$$

Question 2: Support Vector Machine (SVM) — Primal, Lagrangian, and Dual (Optional)

Consider a binary classification dataset $\{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$ and $y_i \in \{+1, -1\}$.

- Write the *hard-margin SVM* primal optimization problem.
- (Lagrangian & stationarity) Form the Lagrangian with multipliers $\alpha_i \geq 0$, derive $\nabla_w \mathcal{L} = 0$ and $\partial \mathcal{L} / \partial b = 0$.
- (Dual) Derive the dual problem (objective and constraints in α only).
- (KKT & support vectors) State KKT complementary slackness and explain support vectors. Explain how to recover (w, b) from α^* .
- (Soft margin) Give the *soft-margin* primal with slack variables and $C > 0$, state its dual, and the bounds on α_i .
- (Kernelization) Explain how the dual enables kernels and write the kernelized decision function.

Answer:

1. Hard-margin primal form.

Seek parameters $w \in \mathbb{R}^d$, $b \in \mathbb{R}$ that maximize the margin:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1, \quad i = 1, \dots, n$$

The objective minimizes $\|w\|$ (equivalently maximizes margin); constraints enforce correct classification with margin at least 1.

2. Lagrangian and stationarity.

Introduce dual variables $\alpha_i \geq 0$ for each inequality; form the Lagrangian (primal is minimization, constraints written as $g_i(w, b) = y_i(w^\top x_i + b) - 1 \geq 0$):

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i (y_i(w^\top x_i + b) - 1).$$

Stationarity (set partial derivatives w.r.t. primal variables to zero): Δ

- w.r.t. w :

$$\frac{\partial \mathcal{L}}{\partial w} = w - \sum_{i=1}^n \alpha_i y_i x_i = 0 \quad \Rightarrow \quad w = \sum_{i=1}^n \alpha_i y_i x_i.$$

- w.r.t. b :

$$\frac{\partial \mathcal{L}}{\partial b} = - \sum_{i=1}^n \alpha_i y_i = 0 \quad \Rightarrow \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

3. Dual problem.

Substitute the stationarity relations into $\mathcal{L}$ to eliminate w, b . Using $w^\top w = (\sum_i \alpha_i y_i x_i)^\top (\sum_j \alpha_j y_j x_j) = \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$, one obtains the dual objective (to be maximized in α):

$$\max_{\alpha \in \mathbb{R}^n} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

subject to the linear constraints

$$\alpha_i \geq 0 \quad (i = 1, \dots, n), \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

(This is a convex quadratic program in the α -variables.)

4. KKT conditions and support vectors.

The KKT conditions (necessary and sufficient here due to convexity) comprise:

5. *Primal feasibility*: $y_i(w^\top x_i + b) \geq 1$.
6. *Dual feasibility*: $\alpha_i \geq 0$.
7. *Stationarity*: $w = \sum_i \alpha_i y_i x_i$, $\sum_i \alpha_i y_i = 0$.
8. *Complementary slackness*:

$$\alpha_i (y_i(w^\top x_i + b) - 1) = 0, \quad i = 1, \dots, n.$$

Complementary slackness implies that any index i with $\alpha_i > 0$ must satisfy $y_i(w^\top x_i + b) = 1$ (the constraint is active). Such points lie exactly on the margin and are called *support vectors*. Points with $\alpha_i = 0$ do not contribute to w and are not support vectors.

Recovering the primal solution from an optimal dual α^* :

$$w^* = \sum_{i=1}^n \alpha_i^* y_i x_i.$$

To get b^* pick any support vector x_k with $0 < \alpha_k^*$ and use

$$b^* = y_k - (w^*)^\top x_k.$$

(Averaging $y_k - (w^*)^\top x_k$ over several margin support vectors improves numerical stability.)

9. Soft-margin SVM (slack variables).

Introduce slack variables $\xi_i \geq 0$ and penalty parameter $C > 0$. Primal:

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (i = 1, \dots, n)$$

Form the Lagrangian with multipliers $\alpha_i \geq 0$ for the first constraint and $\beta_i \geq 0$ for $\xi_i \geq 0$. Eliminating w, b, ξ leads to the dual

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \quad \text{subject to} \quad 0 \leq \alpha_i \leq C, \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

Compared to the hard-margin case the only change in bounds is the upper bound $\alpha_i \leq C$ (coming from the slack penalty). Complementary slackness now also includes terms $\beta_i \xi_i = 0$ and $\alpha_i (y_i (w^\top x_i + b) - 1 + \xi_i) = 0$.

10. Kernelization and decision function.

The dual depends on the data only through inner products $\langle x_i, x_j \rangle$. Replacing these by a kernel function $k(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$ (implicit feature map Φ) yields a kernelized dual program (same form, with dot products $\rightarrow$ k -values). The resulting decision function for a test point x is

$$f(x) = \sum_{i=1}^n \alpha_i^* y_i k(x_i, x) + b^*, \quad \text{predict sign}(f(x)).$$

Only support vectors (indices with $\alpha_i^* > 0$) appear in the sum, so the classifier evaluates a weighted combination of kernel similarities to the support vectors.

Question 3: Bilevel Optimization

Introduction (short). Bilevel optimization models a leader choosing variables anticipating the follower's optimal response. Below is a simple scalar example solvable analytically:

The leader chooses $x \in \mathbb{R}$ to

$$\min_x F(x, y) = (x - 2)^2 + (y - 3)^2 \quad \text{s.t. } x \geq 0,$$

while the follower, given x , chooses $y \in \mathbb{R}$ to

$$\min_y f(x, y) = (y - x^2)^2 + y \quad \text{s.t. } y \geq 0.$$

- Write the bilevel formulation (upper and lower).
- For fixed x , solve the follower and obtain $y^*(x)$.
- Substitute $y^*(x)$ into the leader objective and reduce to one variable.
- Find (x^*, y^*) .
- Discuss regularity (convexity of follower, uniqueness, validity of substitution).

Answer:

1. Bilevel formulation (compact).

Leader (upper level):

$$\min_{x \in \mathbb{R}} F(x, y) = (x - 2)^2 + (y - 3)^2, \quad x \geq 0,$$

with the constraint that the follower selects

$$y \in \arg \min_{y \geq 0} f(x, y) = (y - x^2)^2 + y.$$

2. Follower problem (fixed x).

Define $g(y) = (y - x^2)^2 + y$. Then

$$g'(y) = 2(y - x^2) + 1 = 2y - 2x^2 + 1, \quad g''(y) = 2 > 0,$$

so g is strictly convex in y . The unconstrained stationary point solves

$$2y - 2x^2 + 1 = 0 \implies y_{\text{uncon}} = x^2 - \frac{1}{2}.$$

Enforce the constraint $y \geq 0$ by projection onto $[0, \infty)$. Thus

$$y^*(x) = \max\left(0, x^2 - \frac{1}{2}\right).$$

Note the switching threshold at $x = \frac{1}{\sqrt{2}} \approx 0.7071$. Since $g''(y) > 0$, this minimizer is unique for every fixed x .

3. Reduced leader objective $\Phi(x) = F(x, y^*(x))$.

We consider two regimes:

4. *Region A*: $0 \leq x \leq \frac{1}{\sqrt{2}}$. Here $y^*(x) = 0$, so

$$\Phi_A(x) = (x - 2)^2 + 9.$$

This is a convex quadratic; its unconstrained minimizer is $x = 2$ (outside the region), hence the minimum on the interval occurs at the right endpoint $x = 1/\sqrt{2}$:

$$\Phi_A(1/\sqrt{2}) = \left(\frac{1}{\sqrt{2}} - 2\right)^2 + 9 \approx 10.6716.$$

5. *Region B*: $x \geq \frac{1}{\sqrt{2}}$. Here $y^*(x) = x^2 - \frac{1}{2}$ and

$$\Phi_B(x) = (x - 2)^2 + \left(x^2 - \frac{1}{2} - 3\right)^2 = (x - 2)^2 + (x^2 - 3.5)^2.$$

Differentiate to find stationary points:

$$\Phi'_B(x) = 2(x - 2) + 4x(x^2 - 3.5).$$

Setting $\Phi'_B(x) = 0$ yields the cubic

$$x^3 - 3x - 1 = 0.$$

Its real roots are approximately -1.5321 , -0.3473 , $+1.87938524$; only the last lies in Region B.

6. Optimal pair and objective value.

Pick the feasible root in Region B:

$$x^* \approx 1.8793852415718.$$

Then

$$y^* = x^{*2} - \frac{1}{2} \approx 3.032088862379,$$

and the leader objective evaluates to

$$\Phi(x^*) \approx 0.0155776165707.$$

Compactly:

$$x^* \approx 1.87938524, \quad y^* \approx 3.03208889, \quad \Phi(x^*) \approx 0.0155776$$

This value is much smaller than the best value in Region A (≈ 10.67), hence the displayed pair is the global minimizer for $x \geq 0$.

7. Regularity, uniqueness, and substitution validity (short).

8. The follower objective $g(y) = (y - x^2)^2 + y$ is strictly convex in y (second derivative = 2), and the feasible set $[0, \infty)$ is convex. Therefore for each fixed x the follower has a unique global minimizer $y^*(x)$.
9. Uniqueness and (piecewise) continuity of $y^*(x)$ justify substituting $y^*(x)$ into the upper-level objective to obtain the reduced single-variable program $\min_{x \geq 0} \Phi(x)$.
10. Because $y^*(x)$ is piecewise-smooth (projection causes a kink at $x = 1/\sqrt{2}$), the reduction requires examining the separate regions and comparing boundary values — precisely the approach used above.
11. *Caveat:* if the follower problem were nonconvex or had multiple minimizers, substitution without care can be invalid; then alternative MPEC/KKT reformulations or equilibrium constraints must be used.

Question 4: Multi-Objective Optimization — Pareto Front (Optional)

Consider $(x, y) \in \mathbb{R}^2$ with objectives

$$f_1(x, y) = (x - 2)^2 + y^2, \quad f_2(x, y) = x^2 + (y - 3)^2.$$

(a) Define Pareto optimality for (x^*, y^*) .

(b) Weighted-sum scalarization: for $w \in [0, 1]$ set

$$F_w(x, y) = w f_1(x, y) + (1 - w) f_2(x, y).$$

1. Derive $(x(w), y(w))$ minimizing F_w .
2. Show that as w varies, these trace the Pareto set.
- (c) Give a parametric expression for the Pareto front $\mathcal{P} = \{(f_1(x(w), y(w)), f_2(x(w), y(w))) : w \in [0, 1]\}$ and a closed relation between $u = f_1$ and $v = f_2$.
- (d) (Epsilon-constraint and KKT) Minimize f_1 subject to $f_2 \leq t$; relate the multiplier λ to w and conclude how this maps to a Pareto point.
- (e) Compute $(x(w), y(w))$ and (f_1, f_2) for $w \in \{0, 0.25, 0.5, 0.75, 1\}$ (rounded to 4 decimals).

Answer:

1. Pareto optimality

A point (x^*, y^*) is Pareto optimal if there is no other (x, y) with $f_i(x, y) \leq f_i(x^*, y^*)$ for $i = 1, 2$ and at least one strict inequality. Equivalently: you cannot improve one objective without worsening the other.

2. Weighted-sum scalarization and minimizer.

Form

$$F_w(x, y) = w f_1(x, y) + (1 - w) f_2(x, y), \quad w \in [0, 1],$$

where $f_1(x, y) = (x - 2)^2 + y^2$ and $f_2(x, y) = x^2 + (y - 3)^2$. Expand and collect terms:

$$F_w(x, y) = x^2 + y^2 - 4wx - 6(1 - w)y + (4w + 9(1 - w)).$$

The gradient vanishes at

$$2x - 4w = 0, \quad 2y - 6(1 - w) = 0,$$

hence the unique minimizer (global, since F_w is strictly convex) is

$$(x(w), y(w)) = (2w, 3(1 - w)), \quad w \in [0, 1].$$

Why these are Pareto points. For $w \in (0, 1)$ the weighted-sum minimizer is Pareto optimal for convex problems. As w varies from 0 to 1, $(x(w), y(w))$ traces the straight segment from $(0, 3)$ to $(2, 0)$, which is the Pareto set in decision space.

3. Pareto front (objective space).

Evaluate objectives on the parametric solution:

$$f_1(w) = (2w - 2)^2 + (3(1 - w))^2 = 13(1 - w)^2, \quad f_2(w) = (2w)^2 + (3(1 - w) - 3)^2 = 13w^2.$$

Thus the Pareto front parametric form is

$$(u(w), v(w)) = (13(1 - w)^2, 13w^2), \quad w \in [0, 1].$$

Eliminating w gives a simple closed relation: with $u = f_1$, $v = f_2$,

$$\sqrt{\frac{u}{13}} + \sqrt{\frac{v}{13}} = 1 \iff \sqrt{u} + \sqrt{v} = \sqrt{13}, \quad u \geq 0, v \geq 0.$$

4. Epsilon-constraint and KKT $\leftrightarrow$ weighted sum.

Consider $\min f_1(x, y)$ s.t. $f_2(x, y) \leq t$. Lagrangian with multiplier $\lambda \geq 0$:

$$\mathcal{L} = f_1(x, y) + \lambda(f_2(x, y) - t).$$

Stationarity gives

$$2(x - 2) + 2\lambda x = 0, \quad 2y + 2\lambda(y - 3) = 0.$$

Rearranged:

$$x(1 + \lambda) = 2, \quad y(1 + \lambda) = 3\lambda.$$

Define $w := \frac{1}{1+\lambda} \in (0, 1]$ (when $\lambda \in [0, \infty)$). Then

$$x = 2w, \quad y = 3(1 - w),$$

which matches the weighted-sum solution with parameter w . Thus the KKT multiplier relates to the weight via

$$w = \frac{1}{1 + \lambda}.$$

When the epsilon constraint is active ($\lambda > 0$) one obtains interior Pareto points ($w \in (0, 1)$); when inactive ($\lambda = 0$) one recovers $w = 1$ (pure f_1 minimization).

5. Numerical examples and quick plot snippet.

For $w \in \{0, 0.25, 0.5, 0.75, 1\}$ we have $(x, y) = (2w, 3(1 - w))$ and $(f_1, f_2) = (13(1 - w)^2, 13w^2)$. Rounded to 4 decimals:

w	$(x(w), y(w))$	f_1	f_2
0	(0.0000, 3.0000)	13.0000	0.0000
0.25	(0.5000, 2.2500)	7.3125	0.8125
0.5	(1.0000, 1.5000)	3.2500	3.2500
0.75	(1.5000, 0.7500)	0.8125	7.3125
1	(2.0000, 0.0000)	0.0000	13.0000

Question 5: Batch vs. Stochastic Gradient Descent

Compare Batch Gradient Descent (BGD) and Stochastic Gradient Descent (SGD). Discuss:

- (a) Computational load per iteration.
- (b) Noise (variance) in the gradient estimation.
- (c) Stability and nature of convergence to the minimum.

Answer

- (a) Computational load.

- (b) **BGD**: Each iteration uses the *entire* dataset. With N samples and d features, one step costs $O(N \cdot d)$ — expensive for large N .
- (c) **SGD**: Each iteration uses one sample (or a small mini-batch). The per-step cost is $O(d)$, independent of N .
- (d) **Gradient noise**.
- (e) **BGD**: Uses the exact gradient of the empirical loss (a zero-variance estimate).
- (f) **SGD**: Uses a noisy, high-variance estimate of the gradient (based on a single sample or mini-batch).
- (g) **Convergence stability**.
- (h) **BGD**: With a proper step size, each step decreases the loss; the path toward a (local) minimum is smooth and stable.
- (i) **SGD**: Noisy updates produce an oscillatory / erratic trajectory; the loss may not decrease at every step. The noise can help escape shallow local minima, but iterations typically oscillate around a minimum rather than converging exactly.

Question 6: Steepest Descent Direction via First-Order Taylor

Gradient Descent uses the direction $p_k = -\nabla f(x_k)$. Using a first-order Taylor approximation, prove this is the *steepest descent* direction in a local neighborhood of x_k . (Hint: For small $\alpha > 0$, minimize $f(x_k + \alpha p)$ over unit vectors $\|p\| = 1$.)

Answer

Consider a small step $s = \alpha p$ with $\|p\| = 1$. By a first-order Taylor expansion about x_k ,

$$f(x_k + \alpha p) \approx f(x_k) + \alpha \nabla f(x_k)^\top p.$$

To obtain the largest instantaneous decrease in f (for small $\alpha > 0$) we minimize the linear term $\nabla f(x_k)^\top p$ over unit vectors p :

$$\min_{\|p\|=1} \nabla f(x_k)^\top p.$$

Write the inner product using the angle θ between $\nabla f(x_k)$ and p :

$$\nabla f(x_k)^\top p = \|\nabla f(x_k)\| \|p\| \cos \theta = \|\nabla f(x_k)\| \cos \theta,$$

since $\|p\| = 1$. The quantity $\cos \theta$ ranges over $[-1, 1]$; its minimum is -1 , attained when $\theta = \pi$, i.e. when p points exactly opposite to $\nabla f(x_k)$. Hence the minimizer is

$$p^* = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}.$$

Therefore the steepest (unit) descent direction at x_k is parallel to $-\nabla f(x_k)$. In practice one often chooses the (un-normalized) search direction

$$p_k = -\nabla f(x_k),$$

which points along the steepest local decrease (the norm of p_k can be absorbed into the step length α).

Summary:

Steepest descent direction: $p^* = -\frac{\nabla f(x_k)}{\ \nabla f(x_k)\ }$, so commonly $p_k = -\nabla f(x_k)$.

Question 7: Pseudoinverse Solution is the Least-Squares Minimizer (Optional)

For an overdetermined system $Ax = b$ with $A \in \mathbb{R}^{m \times n}$ ($m > n$) full column rank, the candidate

$$x^* = (A^\top A)^{-1} A^\top b$$

satisfies the normal equations $A^\top A x^* = A^\top b$. Prove x^* is the *unique global minimizer* of $L(x) = \|Ax - b\|_2^2$.

Answer

Claim. Let $A \in \mathbb{R}^{m \times n}$ with $m > n$ and full column rank, and let

$$x^* = (A^\top A)^{-1} A^\top b.$$

Then x^* is the unique global minimizer of

$$L(x) = \|Ax - b\|_2^2 = (Ax - b)^\top (Ax - b).$$

Proof. Expand the quadratic:

$$L(x) = x^\top A^\top A x - 2b^\top A x + b^\top b.$$

Compute the gradient:

$$\nabla L(x) = 2A^\top (Ax - b).$$

Thus any stationary point satisfies the normal equations

$$A^\top A x = A^\top b.$$

Substituting $x^* = (A^\top A)^{-1} A^\top b$ shows that x^* indeed satisfies the normal equations and hence $\nabla L(x^*) = 0$.

Next examine the Hessian (constant):

$$\nabla^2 L(x) = 2A^\top A.$$

For any nonzero $z \in \mathbb{R}^n$,

$$z^\top (2A^\top A) z = 2(Az)^\top (Az) = 2\|Az\|_2^2.$$

Because A has full column rank, $\ker(A) = \{0\}$, so $Az \neq 0$ whenever $z \neq 0$. Hence $z^\top (2A^\top A) z > 0$ for all $z \neq 0$, i.e. $2A^\top A$ is positive definite.

A positive-definite Hessian implies L is strictly convex. A strictly convex differentiable function has at most one stationary point, and any stationary point is the unique global minimizer. Therefore x^* is the unique global minimizer of $L(x)$.

□

Question 8: Newton and Quasi-Newton Methods (Optional)

Newton's method faces (i) high cost of forming the Hessian and (ii) $O(n^3)$ cost to invert it. Quasi-Newton methods (BFGS/DFP) address these. Answer:

- How is the Newton step computed?
- When can Newton's method fail to converge to a minimum (or converge to a saddle/maximum)?
- When is the Newton direction not a descent direction?
- What is the main idea of Quasi-Newton methods?
- How do they approximate the Hessian (or inverse) without explicit second derivatives? (Secant equation.)

Answer

(a) Newton update / search direction.

With $g_k = \nabla f(x_k)$ and $H_k = \nabla^2 f(x_k)$ the Newton iterate and direction are

$$x_{k+1} = x_k - H_k^{-1} g_k, \quad p_k = -H_k^{-1} g_k.$$

(Equivalently one may solve the linear system $H_k p_k = -g_k$ for p_k rather than forming the inverse.)

(b) Failure modes for (pure) Newton's method.

- (c) H_k singular (non-invertible) so H_k^{-1} does not exist.
- (d) H_k not positive definite (e.g. near saddles or maxima, or far from the minimizer) — direction may not be a descent direction.
- (e) Poor initialization x_0 can lead to divergence or convergence to an undesired critical point.

(f) Descent condition and Newton direction.

A direction p_k is a descent direction iff $p_k^\top g_k < 0$. For Newton's direction:

$$p_k^\top g_k = (-H_k^{-1} g_k)^\top g_k = -g_k^\top H_k^{-1} g_k.$$

Thus $p_k^\top g_k < 0$ precisely when $g_k^\top H_k^{-1} g_k > 0$, i.e. when H_k^{-1} is positive definite. Equivalently, $H_k \succ 0$ (positive definite) is sufficient to guarantee the Newton direction is a descent direction. If H_k is indefinite or negative definite, the Newton step may fail to be descending.

(g) Quasi-Newton idea (high level).

Rather than computing or inverting the exact Hessian H_k at every step, quasi-Newton methods maintain and update an approximation $B_k \approx H_k$ (or its inverse approximation $D_k \approx H_k^{-1}$). Updates are chosen to be cheap (rank-one or rank-two) and to incorporate curvature information from recent iterates, avoiding explicit second-derivative evaluations and full matrix inversions while producing directions that (when B_k is positive definite) remain descent directions.

(h) Secant equation and common updates.

Define the step and gradient differences

$$s_k := x_{k+1} - x_k, \quad y_k := g_{k+1} - g_k.$$

The (multi-dimensional) secant condition is

$$B_{k+1} s_k = y_k \quad \text{or equivalently} \quad D_{k+1} y_k = s_k,$$

which generalizes the scalar secant (finite-difference) idea. Practical updates (e.g. BFGS, DFP) are designed to (i) satisfy the secant equation, (ii) be of low rank (cheap to apply/update), and (iii) preserve positive definiteness of the approximation when possible, thereby promoting descent and good local convergence properties.